Identification of a Key Nucleotide Influencing Cas12a crRNA Activity for Universal Photo-controlled CRISPR Diagnostics

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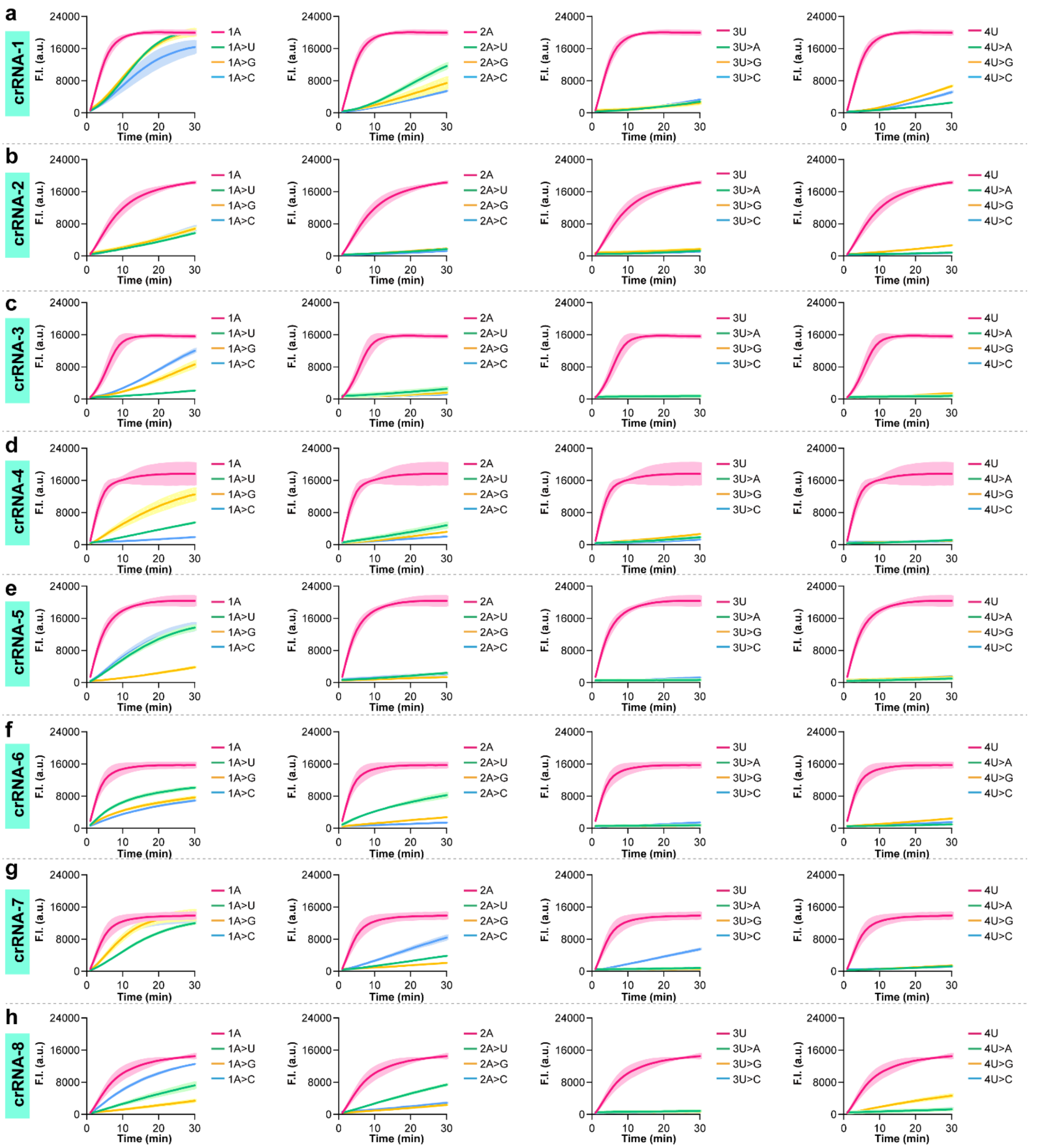
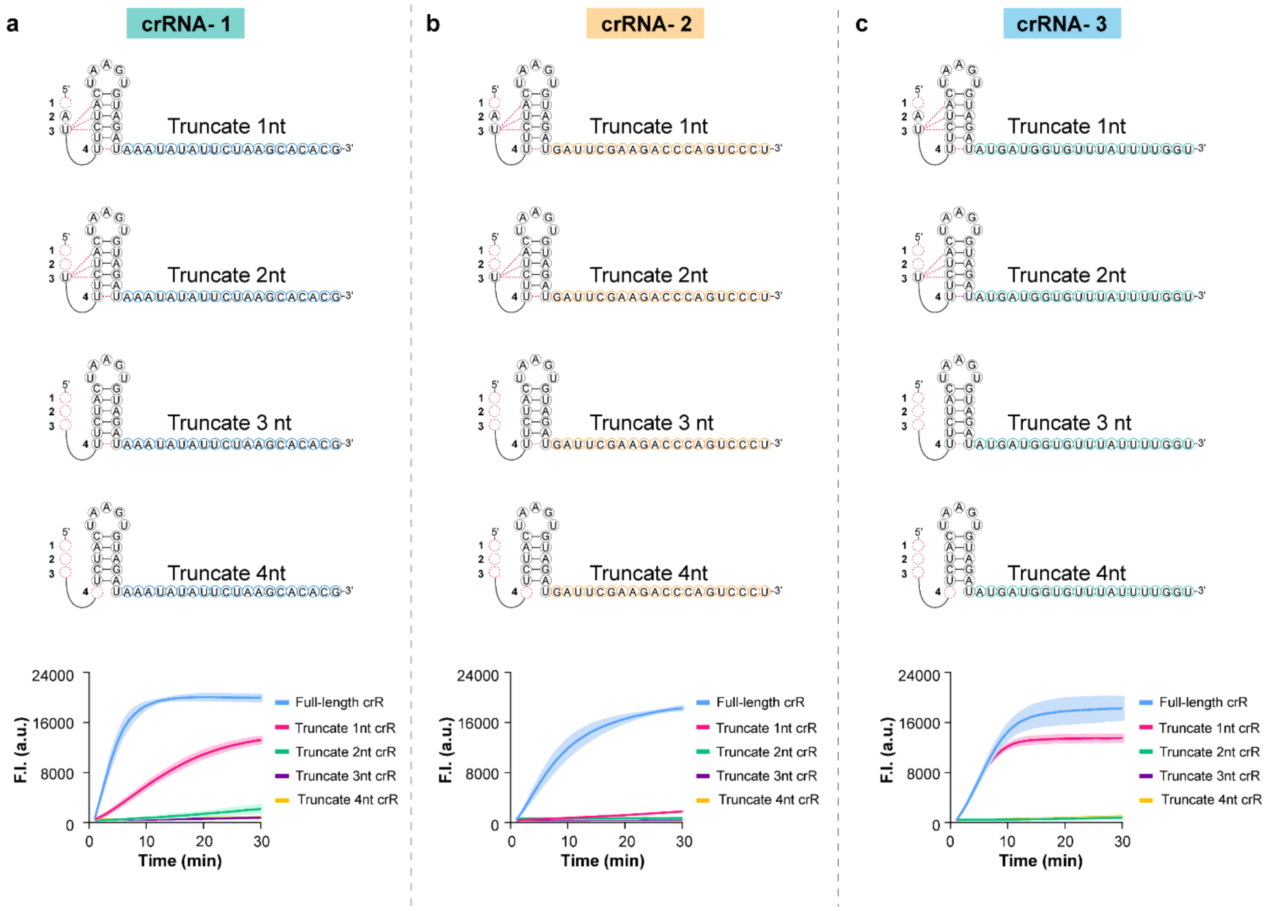
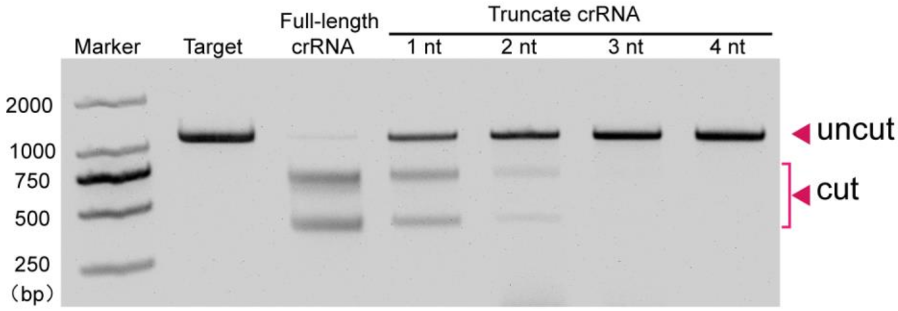


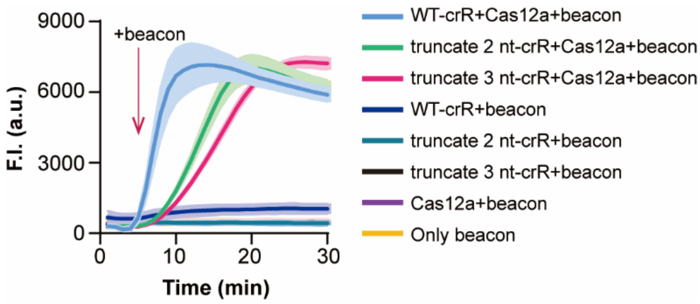
Supplementary Fig.1 | The effect of the *trans*-cleavage activity of crRNA variants with RRS 1-4 sites single-base mutation. a-h crRNA 1-8 represent eight crRNA sequences targeting different targets, respectively. The final concentration of target in each reaction system was 1 nM, and all experiments were conducted in triplicate. Error bars represent the mean values +/- SD (n = 3). Source data are provided as a Source Data file.



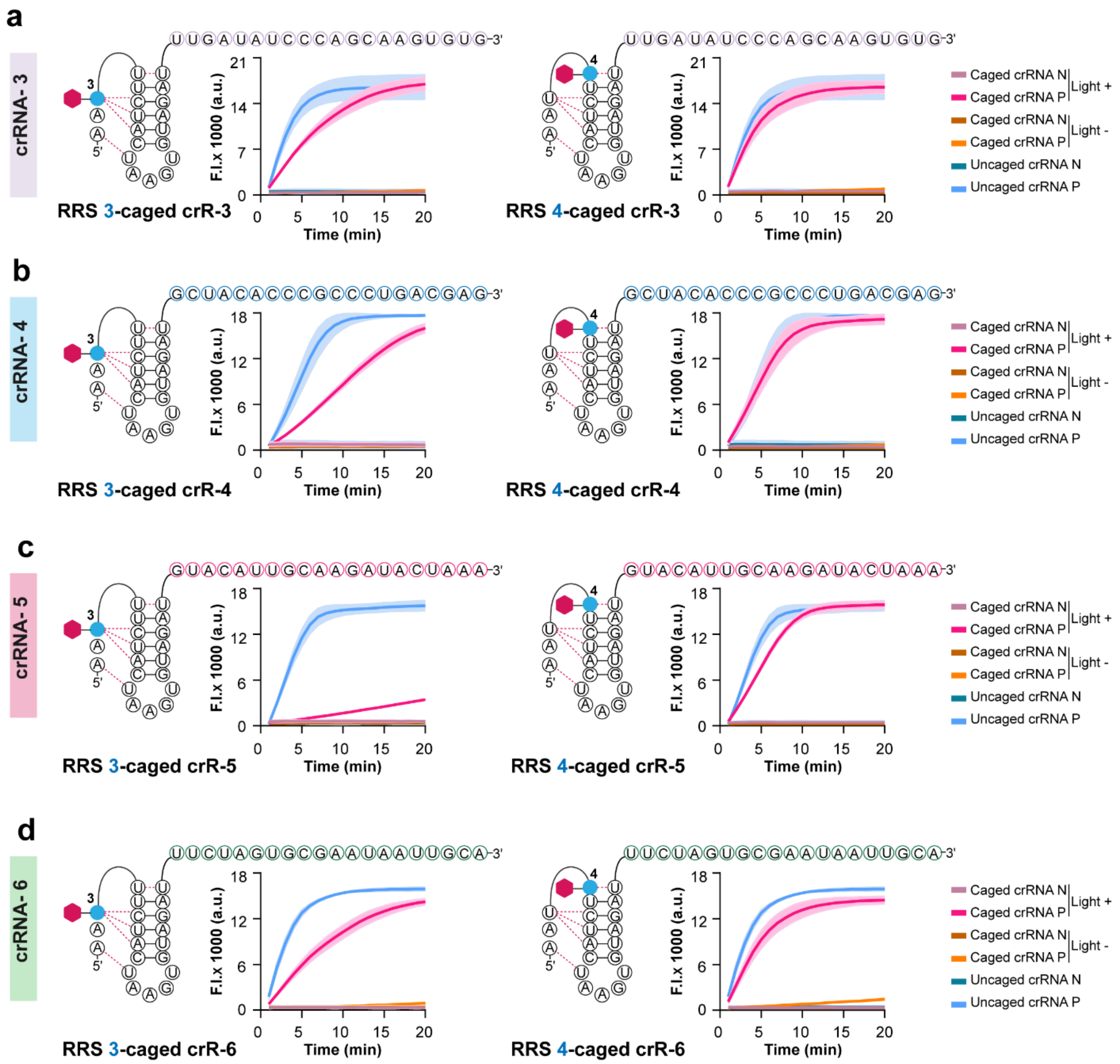
Supplementary Fig.2 | The effect of the *trans*-cleavage activity of crRNA variants with truncation mutation. a-c Schematic diagram of 1-4 nt truncation mutations at the RRS sites and real-time fluorescence curves of *trans*-cleavage activity for the resulting crRNA variants. crRNA 1-3 represent three crRNA sequences targeting different targets, respectively. The final concentration of target in each reaction system was 1 nM, and all experiments were conducted in triplicate. Error bars represent the mean values +/- SD (n = 3). Source data are provided as a Source Data file.



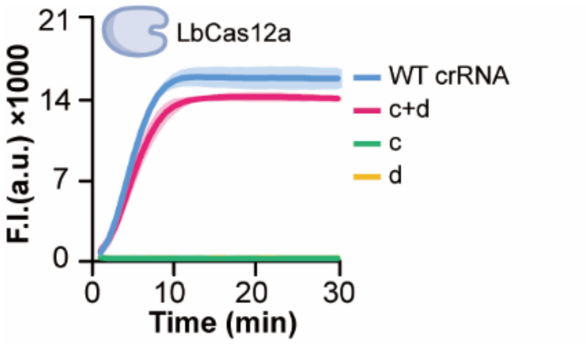
Supplementary Fig.3 ┃ Agarose gel electrophoretic analysis for *cis*-cleavage activity of crRNAs with truncation mutation.



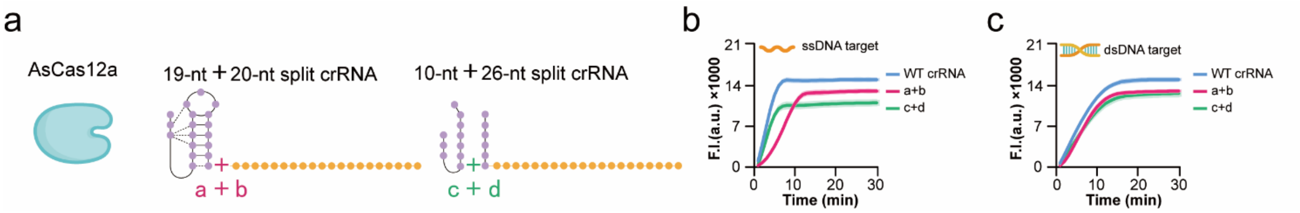
Supplementary Fig.4 | The real-time fluorescence kinetics of Beacon added to the reaction system of crRNA with truncation mutation at the RRS sites. The final concentration of target in each reaction system was 1 nM, and all experiments were conducted in triplicate. Error bars represent the mean values +/- SD (n = 3). Source data are provided as a Source Data file.



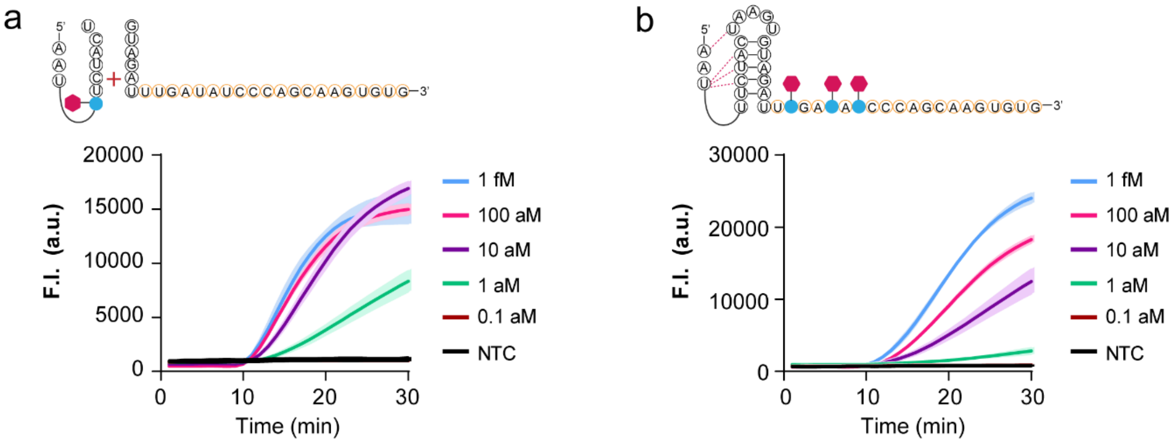
Supplementary Fig.5 | Inactivation and photo-regulated recovery of *trans*-cleavage activity mediated by RRS-3 and RRS-4 single-base photocaged crRNAs. a-d show the sequence diagrams of crRNA3-6 caged at the RRS-3 and RRS-4 sites, respectively, as well as their real-time fluorescence curves for photo-regulated activation and suppression of trans-cleavage activity. The final concentration of target in each reaction system was 1 nM, and all experiments were conducted in triplicate. Error bars represent the mean values +/- SD (n = 3). Source data are provided as a Source Data file. The '+' and '–' symbols denote UV-irradiated and non-irradiated control groups, respectively.



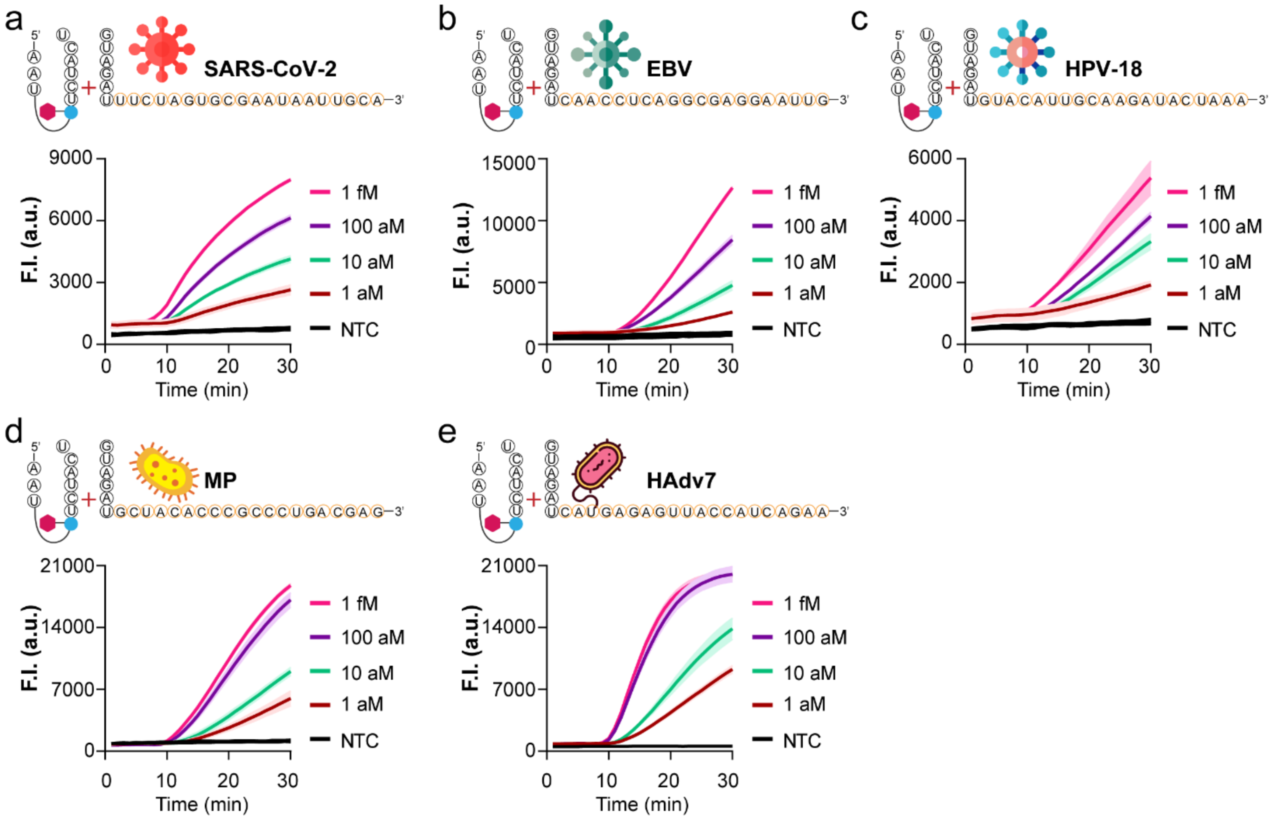
Supplementary Fig.6 | Knockout analysis of the reaction system components for detecting dsDNA targets using LbCas12a stem-loop split crRNA. The final concentration of target in each reaction system was 1 nM, and all experiments were conducted in triplicate. Error bars represent the mean values +/- SD (n = 3). Source data are provided as a Source Data file.



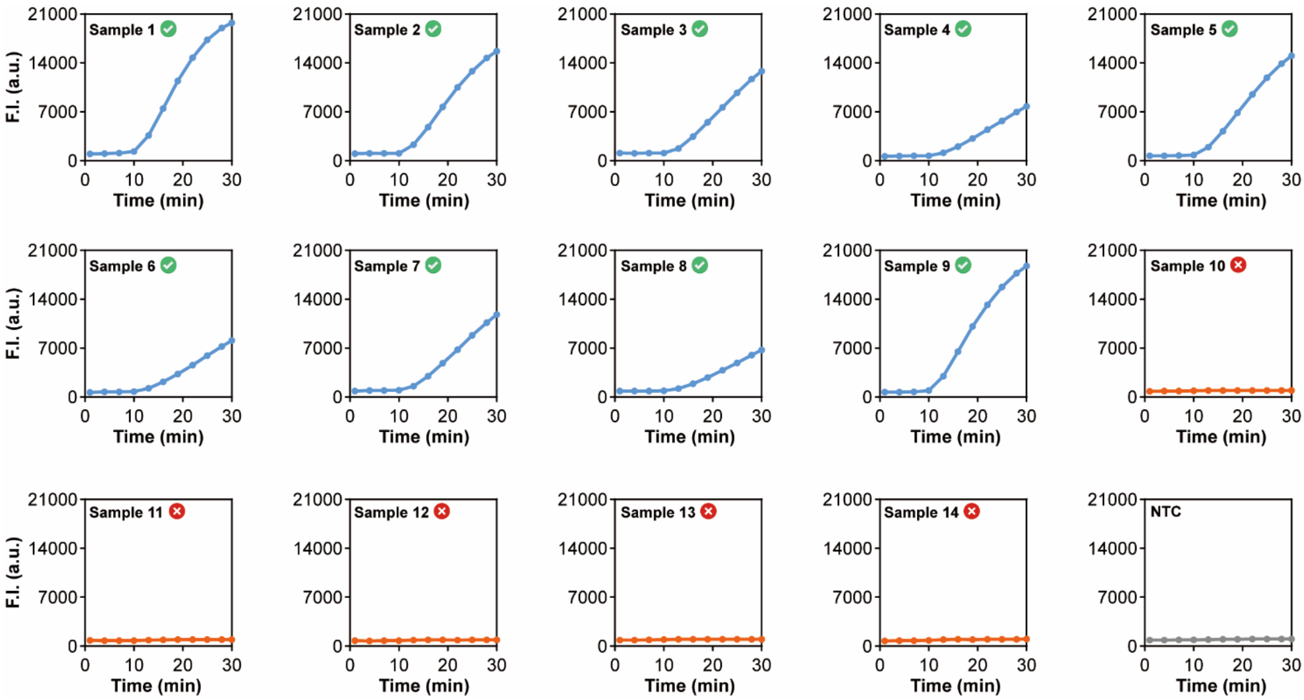
Supplementary Fig.7 | Comparison of the *trans*-cleavage activity between the two crRNA splitting strategies for AsCas12a. a Schematic diagram of the two crRNA splitting strategies for AsCas12a. **b** Real-time fluorescence signal curves of two split crRNA variants for detecting ssDNA targets. The full-length (39-nt) wild-type crRNA was used as a positive control. **c** Real-time fluorescence signal curves of two split crRNA variants for detecting dsDNA targets. The full-length (39-nt) wild-type crRNA was used as a positive control. The final concentration of target in **b** and **c** was 1 nM, and all experiments were conducted in triplicate. Error bars represent the mean values +/- SD (n = 3). Source data are provided as a Source Data file.



Supplementary Fig.8 | Sensitivity analysis of the photo-controlled one-pot system mediated by RRS-4 photocaged split crRNA and spacer region photocaged full-length crRNA. a Real-time fluorescence curve for sensitivity detection using RRS-4 single-base photocaged split crRNA. **b** Real-time fluorescence curve for sensitivity detection using spacer region photo-caged full-length crRNA. Targets were serially diluted with final concentrations of 1 fM, 100 aM, 10 aM, 1 aM and 0.1 aM. All experiments were conducted in triplicate. Error bars represent the mean values +/- SD (n = 3). Source data are provided as a Source Data file.



Supplementary Fig.9 | Universality assessment of the split crRNA-mediated one-pot RPA-LbCas12a photo-controlled system. a Real-time fluorescence curve for sensitivity detection of the SARS-CoV-2 spike gene. **b** Real-time fluorescence curve for sensitivity detection of EBV. **c** Real-time fluorescence curve for sensitivity detection of HPV-18. **d** Real-time fluorescence curve for sensitivity detection of *Mycoplasma pneumoniae* (MP). **e** Real-time fluorescence curve for sensitivity detection detection of *HAdV7*. Targets were serially diluted with final concentrations of 1 fM, 100 aM, 10 aM and 1 aM. All experiments were conducted in triplicate. Error bars represent the mean values +/- SD (n = 3). Source data are provided as a Source Data file.



Supplementary Fig.10 | Real-time fluorescence curves obtained from professional fluorescence quantitative PCR instrument for detecting 14 clinical MP nasopharyngeal swab samples.

Supplementary Table 1. Nucleic acid sequences used in **Fig.S2, Fig.S3 and Fig.S4.**

Name	*Sequence (5'-3')*
Truncate 1 nt-crRNA-1	AUUUCUACUAAGUGUAGAUAAAUAUAUUCUAAGCACACG
Truncate 2 nt-crRNA-1	UUUCUACUAAGUGUAGAUAAAUAUAUUCUAAGCACACG
Truncate 3 nt-crRNA-1	UUCUACUAAGUGUAGAUAAAUAUAUUCUAAGCACACG
Truncate 4 nt-crRNA-1	UCUACUAAGUGUAGAUAAAUAUAUUCUAAGCACACG
Truncate 1 nt-crRNA-2	AUUUCUACUAAGUGUAGAUGAUUCGAAGACCCAGUCCCU
Truncate 2 nt-crRNA-2	UUUCUACUAAGUGUAGAUGAUUCGAAGACCCAGUCCCU
Truncate 3 nt-crRNA-2	UUCUACUAAGUGUAGAUGAUUCGAAGACCCAGUCCCU
Truncate 4 nt-crRNA-2	UCUACUAAGUGUAGAUGAUUCGAAGACCCAGUCCCU
Truncate 1 nt-crRNA-3	AUUUCUACUAAGUGUAGAUAUGAUGGUGUUUAUUUUGCU
Truncate 2 nt-crRNA-3	UUUCUACUAAGUGUAGAUAUGAUGGUGUUUAUUUUGCU
Truncate 3 nt-crRNA-3	UUCUACUAAGUGUAGAUAUGAUGGUGUUUAUUUUGCU
Truncate 4 nt-crRNA-3	UCUACUAAGUGUAGAUAUGAUGGUGUUUAUUUUGCU
Beacon-17nt	TTTTTGGTACTACTTTA
Beacon-BHQ	GATTCGAAGACCCAGTCCCT-BHQ1
Beacon-FAM	FAM-AGGGACTGGGTCTTCGAATCTAAAGTAGTACCAAAAA
FAM-6C-BHQ1	FAM-CCCCCC-BHQ1

Supplementary Table 2. Nucleic acid sequences used in **Fig.2 and Fig.S5.**

Name	*Sequence (5'-3')*
3-caged crR-1	AA**dT***UUCUACUAAGUGUAGAUUUGACUGCCACUGCCGGCAG
4-caged crR-1	AAU**dT***UCUACUAAGUGUAGAUUUGACUGCCACUGCCGGCAG
spacer-caged crR-1	AAUUUCUACUAAGUGUAGAU**UdT***GAC**dT***GCCAC**dT***GCCGGCAG
3-caged crR-2	AA**dT***UUCUACUAAGUGUAGAUCAACCUCAGGCGAGGAAUUG
4-caged crR-2	AAU**dT***UCUACUAAGUGUAGAUCAACCUCAGGCGAGGAAUUG
spacer-caged crR-2	AAUUUCUACUAAGUGUAGAUCAACC**dT***CAGGCGAGGAA**dT*** **dT***G
3-caged crR-3	AA**dT***UUCUACUAAGUGUAGAUUUGAUAUCCCAGCAAGUGUG
4-caged crR-3	AAU**dT***UCUACUAAGUGUAGAUUUGAUAUCCCAGCAAGUGUG
3-caged crR-4	AA**dT***UUCUACUAAGUGUAGAUGCUACACCCGCCCUGACGAG
4-caged crR-4	AAU**dT***UCUACUAAGUGUAGAUGCUACACCCGCCCUGACGAG
3-caged crR-5	AA**dT***UUCUACUAAGUGUAGAUGUACAUUGCAAGAUACUAAA
4-caged crR-5	AAU**dT***UCUACUAAGUGUAGAUGUACAUUGCAAGAUACUAAA
3-caged crR-6	AA**dT***UUCUACUAAGUGUAGAUUUCUAGUGCGAAUAAUUGCA
4-caged crR-6	AAU**dT***UCUACUAAGUGUAGAUUUCUAGUGCGAAUAAUUGCA

Supplementary Table 3. Nucleic acid sequences used in **Fig.3, Fig.S6 and Fig.S7.**

Name	*Sequence (5'-3')*
HPV-18 PCR-F (dsDNA Target)	ATCGCGTCCTTTATCACAG
HPV-18 PCR-R (dsDNA Target)	ATCCTTATTTTCAGCCGGTGCA
HPV-18 ssDNA Target	ACAAATATCCAATGGTACCTCACATTTAGTATCTTGCAATGTACTAAAGTCCATGGCAC
As-WT-crRNA	AAUUUCUACUCUUGUAGAUGUACAUUGCAAGAUACUAAA
Lb-WT-crRNA	AAUUUCUACUAAGUGUAGAUGUACAUUGCAAGAUACUAAA
As-split crR-a	AAUUUCUACUCUUGUAGAU
Lb-split crR-a	AAUUUCUACUAAGUGUAGAU
split crR-b	GUACAUUGCAAGAUACUAAA
split crR-c	AAUUUCUACU
split crR-d	GUAGAUGUACAUUGCAAGAUACUAAA
4-caged crR-c	AAU<u>**dT***</u>UCUACU
4-caged crR-d	GUAGAUUUGAUAUCCCAGCAAGUGUG

Supplementary Table 4. Nucleic acid sequences used in **Fig.4, Fig.5, Fig.S8, Fig.S9 and Fig.S10.**

Name	*Sequence (5'-3')*
MP PCR-F	GCGTGAAGTTCAAGTCTCCTG
MP PCR-R	GCATTGGGATTCCAATGGCA
MP RPA-F1	TTTACCCACCCAGTCACCGATCTGTTTGATC
MP RPA-R1	AGCTCAAGACCTTTAAACGAACCATTTTAGG
MP spacer-caged crRNA1	AAUUUCUACUAAGUGUAGAUU**dT***GA**dT***A**dT***CCCAGCAAGUGUG
MP split crRNA-1	GUAGAUUUGAUAUCCCAGCAAGUGUG
MP RPA-F2	CCACCACGACGCTCAAGCGCCAGCAATTTAG
MP RPA-R2	AGGGAGGAAAAGCTCGTGTTACGATACGTTC
MP split crRNA2	GUAGAUGCUACACCCGCCCUGACGAG
S gene PCR-F	ATGTTTGTTTTTCTTGTTTTATTGCCACTAGTC
S gene PCR-R	TTGTAATATTAGGAAATCTAACAATAGATTCTGTTG
S gene RPA-F	AACAACAAAAGTTGGATGGAAAGTGAGTTCA
S gene RPA-R	CCATAAGAAAAGGCTGAGAGACATATTCAAA
S gene split crRNA	GUAGAUUUCUAGUGCGAAUAAUUGCA
HPV-18 PCR-F (Target)	ATCGCGTCCTTTATCACAG
HPV-18 PCR-R (Target)	ATCCTTATTTTCAGCCGGTGCA
HPV-18 RPA-F	TGGAAGATGGTGATATGGTAGATACTGGATA
HPV-18 RPA-R	CAAATAGACTGACAAATATCCAATGGTACCT
HPV-18 split crRNA	GUAGAUGUACAUUGCAAGAUACUAAA
EBV PCR-F	CACAGATGACCCAGGAGAAGG
EBV PCR-R	TCGCATCCTTCAAAACCTCAG
EBV RPA-F	GAGGAAGGAAATTGGGTCGCCGGTGTGTT
EBV RPA-R	AATGGTGTAATACGACATTGTGGAACAGC
EBV split crRNA	GUAGAUCAACCUCAGGCGAGGAAUUG
HAdv7 PCR-F	GTGCTTGATAGAGGTCCTAGCT
HAdv7 PCR-R	TGTCCTGCAAGTCAACCACT
HAdv7 RPA-F	CAGACAGCCATGTGGTATACAAGCCAGGAA
HAdv7 RPA-R	TAAGACCTACAAAGTTATCCCTGAAGCCAA
HAdv7 split crRNA	GUAGAUCAUGAGAGUUACCAUCAGAA
4-caged crR-c	AAU**dT***UCUACU

Note: **dT*** denotes a deoxyribonucleoside thymine modified with NPOM.

Supplementary Table 5. SARS-CoV-2 S gene plasmid sequence used in this study.

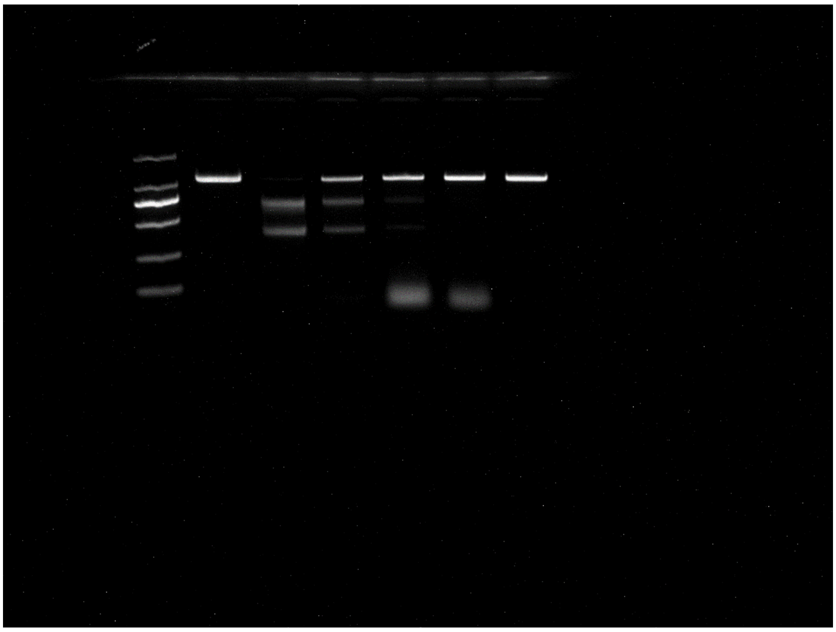
Name	*Sequence (5′-3′)*
SARS-CoV-2 S gene	ATGTTTGTTTTTCTTGTTTTATTGCCACTAGTCTCTAGTCAGTGTGTT AATCTTACAACCAGAACTCAATTACCCCCTGCATACACTAATTCTTTC ACACGTGGTGTTTATTACCCTGACAAAGTTTTCAGATCCTCAGTTTT ACATTCAACTCAGGACTTGTTCTTACCTTTCTTTTCCAATGTTACTTG GTTCCATGCTATACATGTCTCTGGGACCAATGGTACTAAGAGGTTTG ATAACCCTGTCCTACCATTTAATGATGGTGTTTATTTTGCTTCCACTG AGAAGTCTAACATAATAAGAGGCTGGATTTTTGGTACTACTTTAGAT TCGAAGACCCAGTCCCTACTTATTGTTAATAACGCTACTAATGTTGTT ATTAAAGTCTGTGAATTTCAATTTTGTAATGATCCATTTTTGGGTGTT TATTACCACAAAAACAACAAAAGTTGGATGGAAAGTGAGTTCAGAG TTTATTCTAGTGCGAATAATTGCACTTTTGAATATGTCTCTCAGCCTT TTCTTATGGACCTTGAAGGAAAACAGGGTAATTTCAAAAATCTTAGG GAATTTGTGTTTAAGAATATTGATGGTTATTTTAAAATATATTCTAAGC ACACGCCTATTAATTTAGTGCGTGATCTCCCTCAGGGTTTTTCGGCTT TAGAACCATTGGTAGATTTGCCAATAGGTATTAACATCACTAGGTTTC AAACTTTACTTGCTTTACATAGAAGTTATTTGACTCCTGGTGATTCTT CTTCAGGTTGGACAGCTGGTGCTGCAGCTTATTATGTGGGTTATCTT CAACCTAGGACTTTTCTATTAAAATATAATGAAAATGGAACCATTAC AGATGCTGTAGACTGTGCACTTGACCCTCTCTCAGAAACAAAGTGT ACGTTGAAATCCTTCACTGTAGAAAAAGGAATCTATCAAACTTCTAA CTTTAGAGTCCAACCAACAGAATCTATTGTTAGATTTCCTAATATTAC AA

Supplementary Table 6. EBV plasmid sequence used in this study.

Name	*Sequence (5'-3')*
EBV	AGGAGGGGCAGGAGCAGGAGGAGGGGCAGGAGCAGGAGGGGCAG GAGCAGGAGGGGCAGGAGCAGGAGGGGCAGGAGCAGGAGGGGCA GGAGCAGGAGGGGCAGGAGCAGGAGGTGGAGGCCGGGGTCGAGG AGGCAGTGGAGGCCGGGGTCGAGGAGGCAGTGGAGGCCGGGGTC GAGGAGGTAGTGGAGGCCGCCGGGGTAGAGGACGTGAAAGAGCCA GGGGGAGGAGTCGTGAAAGAGCCAGGGGGAGAGGTCGTGGACGT GGTGAAAAGAGGCCCAGGAGTCCCAGTAGTCAGTCATCATCATCCG GGTCTCCACCGCGCAGGCCCCCTCCAGGTAGAAGGCCATTTTTCCA CCCTGTAGGGGATGCCGATTATTTTGAATACCTCCAAGAAGGTGGCC CAGATGGTGAGCCTGACGTGCCCCCGGGAGCGATAGAGCAGGGCC CCACAGATGACCCAGGAGAAGGCCCAAGCACTGGACCCCGGGGTC AGGGTGATGGAGGCAGGCGCAAAAAAGGAGGGTGGTTTGGAAAGC ATCGTGGTCAAGGAGGTTCCAACCCGAAATTTGAGAACATTGCAGA AGGTTTAAGAGTTCTCCTGGCTAGGAGTCACGTAGAAAGGACTACC GAGGAAGGAAATTGGGTCGCCGGTGTGTTCGTATATGGAGGTAGTA AGACCTCCCTTTACAACCTCAGGCGAGGAATTGCCCTTGCTGTTCC ACAATGTCGTATTACACCATTGAGTCGTCTCCCCTTTGGAATGGCCC CTGGACCCGGCCCACAACCTGGCCCACTAAGGGAGTCCATTGTCTG TTATTTCATGGTCTTTTTACAAACTCATATATTTGCTGAGGTTTTGAA GGATGCGATTAAGGACCTTGTTATGACAAAGCCCGCTCCTACCTGCA ATATCAAGGTGACTGTGTGCAGCTTTGACGATGGAGTAGATTTGCCT CCCTGGTTTCCACCTATGGTGGAAGGGGCTGCCGCGGAGGGTGATG ACGGAGATGACGGAGATGAAGGAGGTGATGGAGATGAGGGTGAGG AAGGGCAGGAGTGA

Supplementary Table 7. The *Ct* values of 35 clinical MP samples measured by qPCR.

Sample ID	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
MP (*Ct*)	33.79	27.37	/	34.90	32.67	28.40	33.96	33.99	26.28
Sample ID	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**
MP (*Ct*)	30.61	/	/	30.03	30.83	32.98	23.48	29.46	28.69
Sample ID	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**
MP (*Ct*)	35.06	/	26.75	32.24	29.13	30.42	34.03	27.89	32.69
Sample ID	**28**	**29**	**30**	**31**	**32**	**33**	**34**	**35**	
MP (*Ct*)	28.68	30.49	/	26.17	27.88	34.19	30.54	36.09	



Uncropped scans of agarose gel in Supplementary Fig.3